



Gym Management System (GMS)

- GROUP 1
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 .git	08/11/2025 8:08 CH
 .idea	08/11/2025 8:08 CH
 assets	08/11/2025 8:08 CH
 auth	08/11/2025 8:08 CH
 data	08/11/2025 8:08 CH
 data_work	08/11/2025 8:08 CH
 guis	08/11/2025 8:08 CH
 models	08/11/2025 8:08 CH
 utils	08/11/2025 8:08 CH
 attendance1	06/11/2025 11:11 CH
 doanhthu	06/11/2025 11:11 CH
 GYM MANAGERMENT	06/11/2025 11:11 CH
 main	07/11/2025 1:22 SA

folder .idea

folder assets

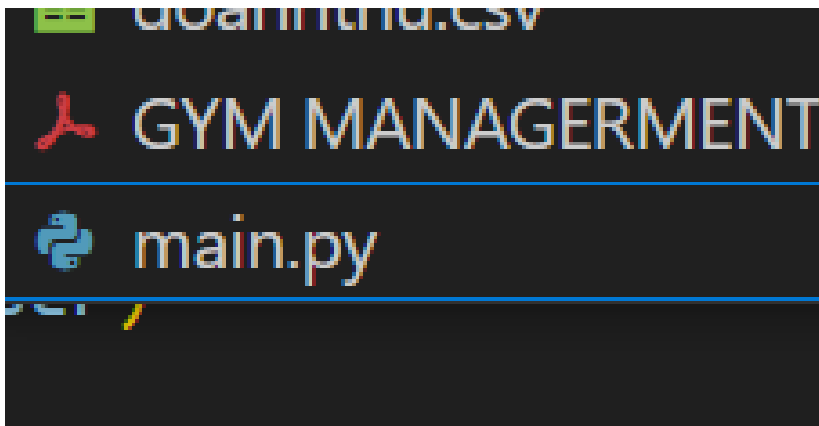
folder auth

folder data

file background_admin

file gym_icon





```
from guis.login_window import open_login_window
from models.admin import Admin
from models.trainer import Trainer
from models.member import Member

from guis.admin_window import open_admin_window
from guis.trainer_window import open_trainer_window
from guis.member_window import open_member_window

def main():

    username, role = open_login_window()

    if role == "admin":
        admin_user = Admin(username)
        open_admin_window(admin_user)

    elif role == "trainer":
        trainer_user = Trainer(username)
        open_trainer_window(trainer_user)

    elif role == "member":
        member_user = Member(username)
        open_member_window(member_user)
        print(f"Member functionality not yet implemented for {username}")

    else:
        print("Login failed.")

if __name__ == "__main__":
    main()
```

assets

auth

data

data_work

guis

models

utils

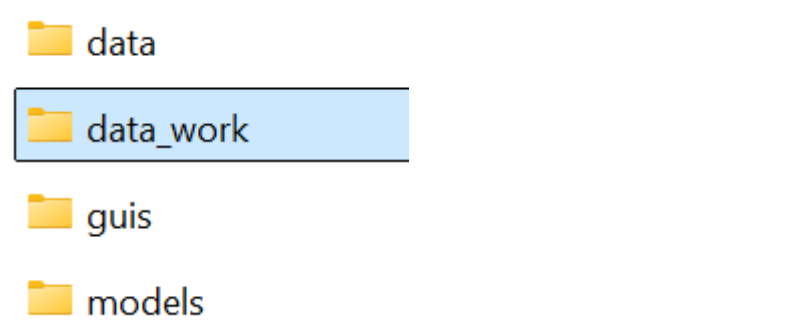
```
{
  "username": "admin_test1",
  "password": "0000000000",
  "role": "admin"
},
{
  "username": "nhathoa",
  "password": "nhathoa",
  "role": "admin"
},
{
  "username": "trainer_test1",
  "password": "1111111111",
  "role": "trainer"
},
{
  "username": "member_test1",
  "password": "2222222222",
  "role": "member"
},
{
  "username": "hoangthong",
  "password": "hoangthong",
  "role": "trainer"
}
```

```
"Dumbbells": 30,
"Rope": 5,
"Balls": 3,
"Chest Press": 6,
"Incline Chest Press": 1,
"Big Barbel Rod": 2,
"Zik-Zak Rod": 7,
"Small Barbel Rod": 4,
"Leg Press Machine": 1,
"Benth": 4,
"T.V": 1,
"Speaker": 2,
```

```
{
  "id": 1,
  "f_name": "John",
  "l_name": "Doe",
  "gender": "male",
  "phone": "1234567890",
  "date_of_birth": "1990-01-01",
  "joined_date": "2023-01-01",
  "address": "123 Main St, City, Country",
  "email": "john@example.com",
  "username": "johndoe",
  "height": "180",
  "weight": "80",
  "trainer_id": "1",
  "member_username": [
    "Dat Kim",
    "Cuc Bach",
    "aaa"
  ]
}
```

```
{
  "username": "datkim",
  "attendance_dates": [
    "2025-07-01",
    "2025-07-02",
    "2025-07-03",
    "2025-07-04",
    "2025-07-05",
    "2025-10-22",
    "2025-11-04",
    "2025-11-07"
  ],
  "trainer_id": "1",
  "member_username": "5"
```

```
{
  "1 Months": [
    30,
    1000000
  ],
  "3 Months": [
    90,
    2800000
  ],
  "6 Months": [
    180,
    5500000
  ],
  "12 Months": [
    365,
    10000000
  ],
  "36 Months": [
    1200,
    25000000
  ],
  "Lifetime": [
    400000,
    100000000
  ],
  "a": [
    123,
    123
  ]
}
```



```
import json
import os

def load_users_data():
    if not os.path.exists("data/user.json"):
        return []
    with open('data/user.json', 'r') as f:
        return json.load(f)

def check_credentials(username=None, password=None):
    users_information = load_users_data()

    for user in users_information:
        if user["username"] == username and user["password"] == password:
            print("Login successful!")
            return user["role"]

    return None

def add_user_to_json(username, password, role="member"):
    if not os.path.exists("data/user.json"):
        return []
    with open('data/user.json', 'r') as f:
        users = json.load(f)

    users.append({"username": username, "password": password, "role": role})

    with open('data/user.json', "w",) as f:
        json.dump(users, f, indent=4)
```

> _pycach
auth_add
info_add

```
def change_equipment_in_json(equipment_data):
    filepath = "data/equipment_info.json"
    with open(filepath, "w", encoding="utf-8") as f:
        json.dump(equipment_data, f, indent=4)

def delete_user_from_json(username):

    # Delete from user.json
    user_filepath = "data/user.json"
    try:
        with open(user_filepath, "r", encoding="utf-8") as f:
            users = json.load(f)
        # Filter out the user with matching username
        users = [user for user in users if user.get("username") != username]
        with open(user_filepath, "w", encoding="utf-8") as f:
            json.dump(users, f, indent=4)
    except (FileNotFoundError, json.JSONDecodeError) as e:
        print(f"Error reading user.json: {e}")
        return
```

```
def load_users_data():
    try:
        if not os.path.exists("data/member_info.json"):
            return []
        with open("data/member_info.json", "r", encoding="utf-8") as f:
            return json.load(f)
    except (FileNotFoundError, json.JSONDecodeError, Exception) as e:
        print(f"Error loading users data: {e}")
        return []

def load_trainers_data():
    if not os.path.exists("data/trainer_info.json"):
        return []
    with open('data/trainer_info.json', 'r') as f:
        return json.load(f)
```


▼ guis

- > __pycache__
- > admin_guis
- > member_guis
- > trainer_guis
- 🔗 admin_window.py
- 🔗 login_window.py

```
def open_admin_window(admin):  
    root = tk.Tk()  
    root.title("Gym Management System - /  
  
# --- GỌI BACKGROUND CỦA BẠN LÊN TRÊN ĐẦU ---  
c_icon(root)  
c_background(root)
```

```
root.state('zoomed')
```

```
# --- Cấu hình Kiểu dáng (Styling) ---  
style = ttk.Style()  
style.theme_use('clam')
```

```
# Màu sắc
```

```
sidebar_bg = █ "#2c3e50"  
button_fg = █ "#ffffff"  
button_hover_bg = █ "#34495e"
```

▼ admin_guis

- 🔗 add_attendance.py
- 🔗 cal_revenue.py
- 🔗 delete_member.py
- 🔗 equipment_manager.py
- 🔗 manage_subscription.py
- 🔗 new_member.py
- 🔗 new_trainer.py
- 🔗 search_member.py
- 🔗 search_trainer.py
- 🔗 top_members.py

▼ member_guis

- 🔗 check_info.py
- 🔗 check_workout.py
- 🔗 contact.py
- 🔗 renew_subscription.py
- 🔗 track_progress.py
- 🔗 view_subscription.py
- > trainer_guis
- 🔗 admin_window.py
- 🔗 login_window.py
- 🔗 member_window.py
- 🔗 register_window.py

```
def open_login_window():
```

```
    result = {}
```

```
    root = tk.Tk()
```

```
    root.geometry("800x600")
```

```
    c_icon(root)
```

```
    root.title("Gym Management System - Login")
```

```
    root.configure(bg=█ "#f0f2f5")
```

▼ models

> __pycache__

🐍 admin.py

🐍 member.py

🐍 trainer.py

🐍 user.py

```
def save_equipment_data(self, equipment_data):
    try:
        change_equipment_in_json(equipment_data)
        return True
    except Exception as e:
        print(f"Error saving equipment data: {e}")
        return False
```

```
def delete_member(self, username):
    try:
        delete_user_from_json(username)
        return True
    except Exception as e:
        print(f"Error deleting member: {e}")
        return False
```

```
def search_members(self, search_term):
    try:
        return search_members_by_name(search_term)
    except Exception as e:
        print(f"Error searching members: {e}")
        return []
```

```
def search_trainer(self, search_term):
    try:
        return search_trainer_by_name(search_term)
    except Exception as e:
        print(f"Error searching trainers: {e}")
        return []
```


> models

✓ utils

> __pycache__

🐍 create_background.py

🐍 create_icon.py

```
import tkinter as tk
from PIL import Image, ImageTk
```

```
def c_background(window: tk.Tk):
```

```
    # Load and resize the background image
    bg_image = Image.open("assets/background_admin.png")
    #bg_image = Image.open("assets/gym_icon.png")
    bg_image = bg_image.resize((window.winfo_screenwidth(),window.winfo_screenheight()))
    bg_photo = ImageTk.PhotoImage(bg_image)

    # Set image as a label background
    bg_label = tk.Label(window, image=bg_photo)
    bg_label.place(x=0, y=0, relwidth=1, relheight=1)

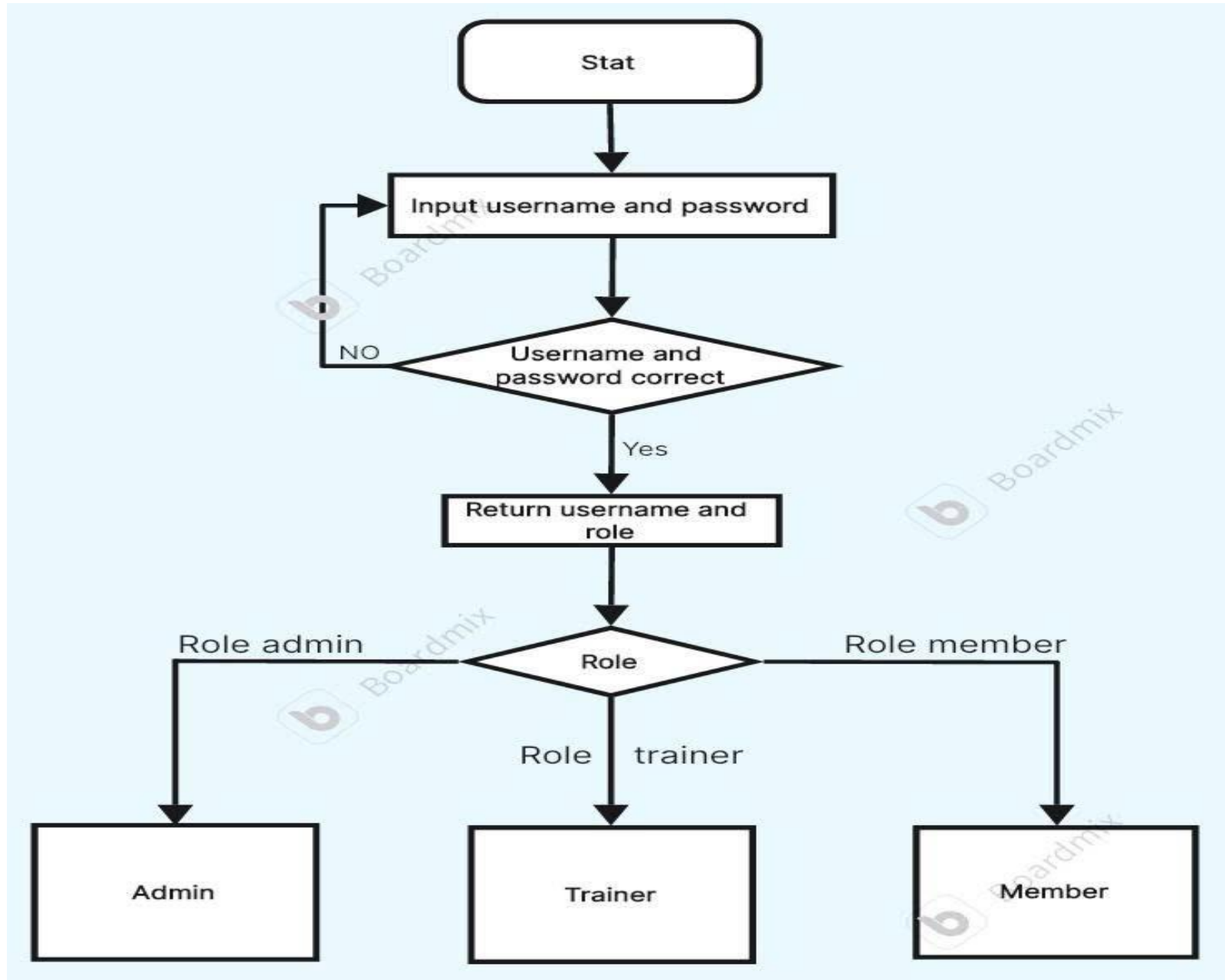
    bg_label.image = bg_photo
```

```
from tkinter import PhotoImage, Tk
```

```
def c_icon(window: Tk ):
```

```
    photo = PhotoImage(file = "assets/gym_icon.png") # Placeholder for the logo path
    window.iconphoto(True, photo) #Set the icon for the main window
```

LOGIN INTERFACE FLOWCHART



LOGIN WINDOW

 Gym Management System - Login—□×

WELCOME TO THE GYM

Username:

Password:

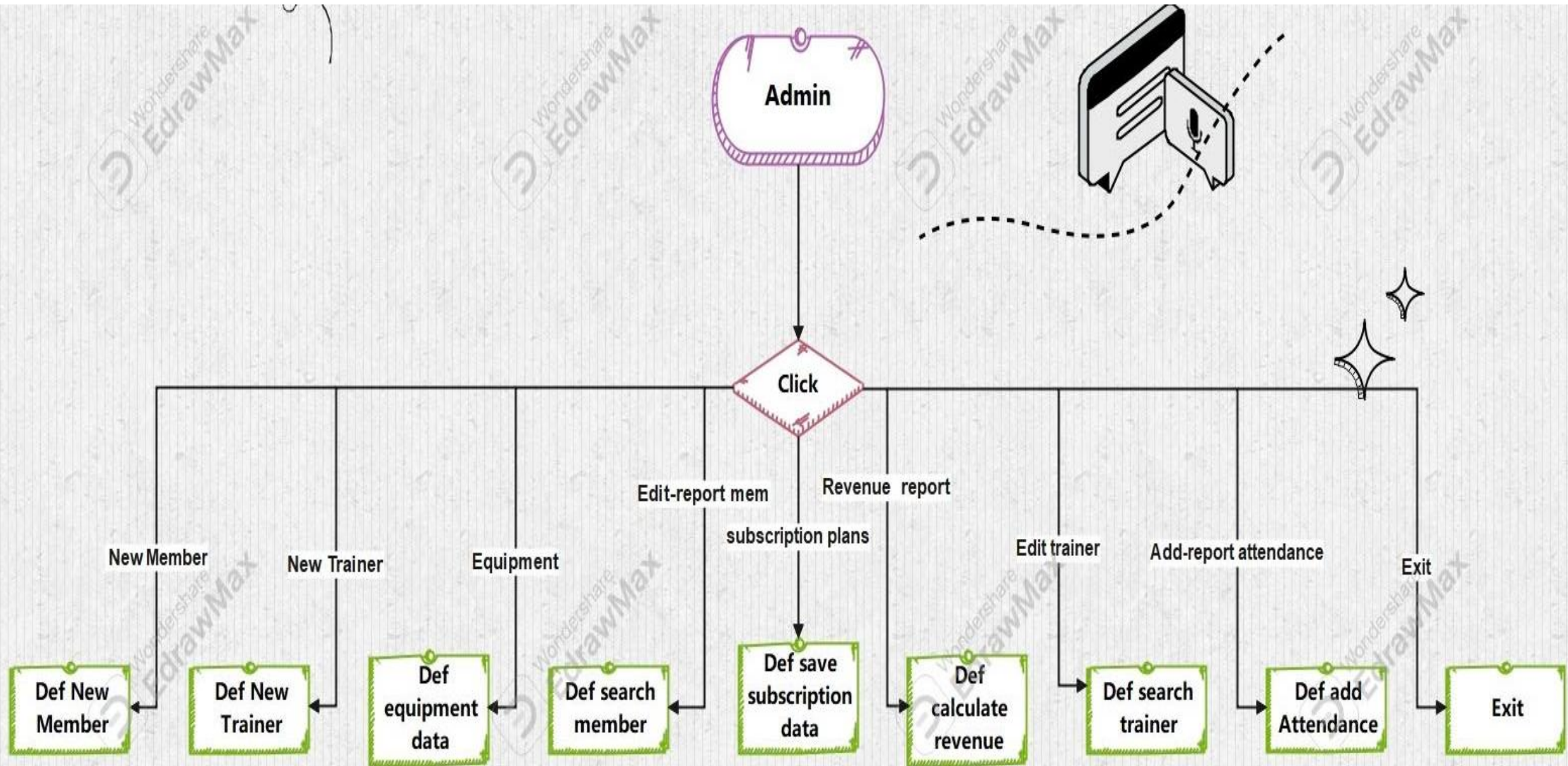
☐ Show Password

Login

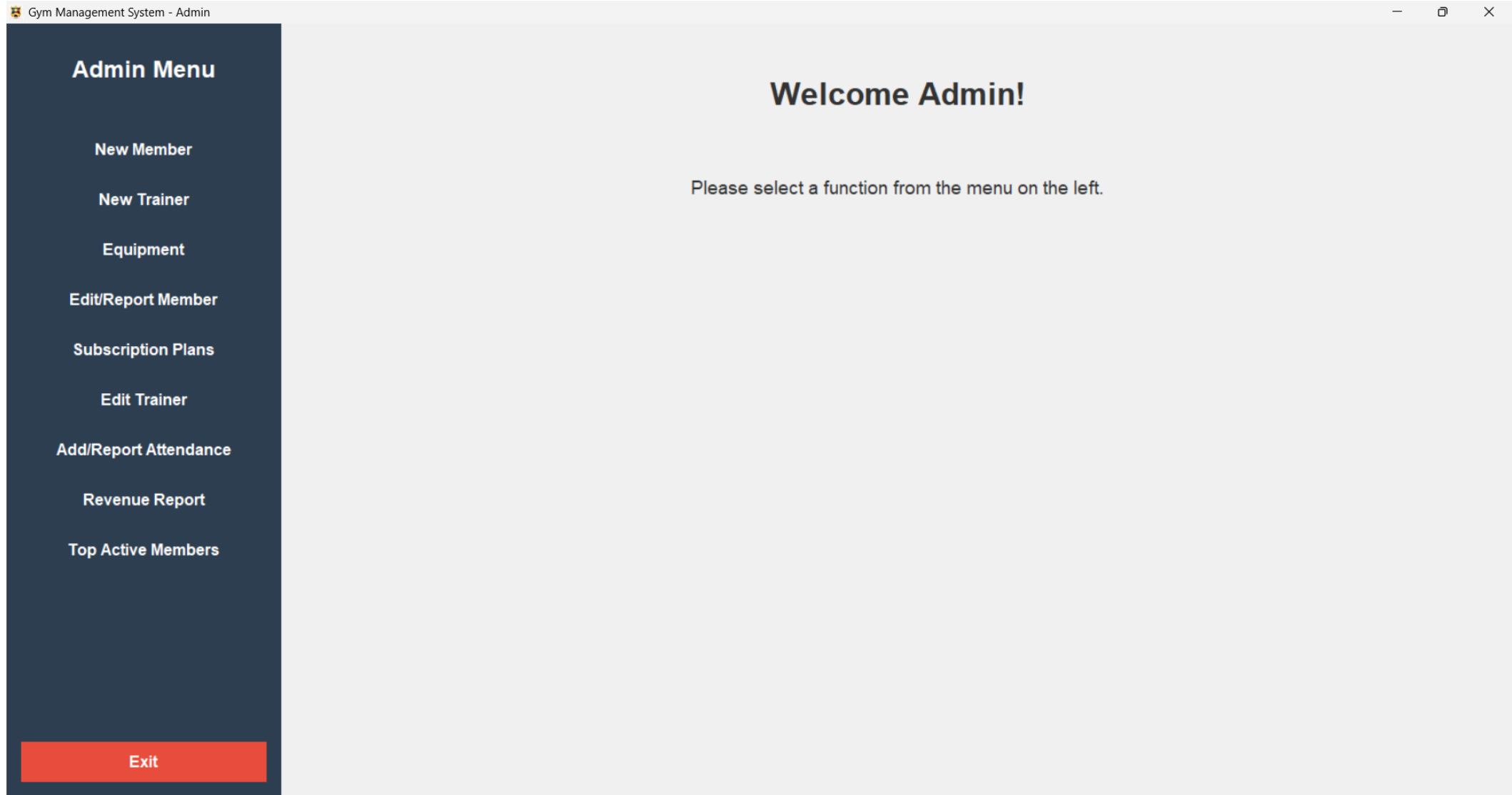
Don't have an account?

[Register](#)

Gym Management system Functions (Admin)



ADMIN WINDOW



Some of the GUI in Admin

New Trainer

First Name	
Last Name	
Phone	
Address	
Email	
Username	
Height (cm)	
Weight (kg)	
Trainer ID	
Gender	☒ Male ☐ Female
Date of Birth	2000-01-01
Joined Date	2025-11-10

Submit

Back to Menu

New Member

First Name	
Last Name	
Phone	
Address	
Email	
Username	
Gender	☒ Male ☐ Female
Date of Birth	2000-01-01
Joined Date	2025-11-10
Membership	1 Months
Trainer Username	johndoe
Subscription Expiry	2025-12-10

Submit

Back to Menu

Equipment Manager

Dumbbells	30	+	-	Benth	4	+	-
Rope	5	+	-	T.V	1	+	-
Balls	3	+	-	Speeker	2	+	-
Chest Press	6	+	-	2 kg Weight	8	+	-
Incline Chest Press	1	+	-	5 kg Weight	5	+	-
Big Barbel Rod	2	+	-	10 kg Weight	6	+	-
Zik-Zak Rod	7	+	-	20 kg Weight	2	+	-
Small Barbel Rod	4	+	-	30 kg Weight	3	+	-
Leg Press Machine	1	+	-	50 kg Weight	4	+	-

Save

Back to Menu

Some of the GUI in Admin

Search and Edit Member

Search by Name:

Username	First Name	Last Name	Email	Phone	Membership	Expiry
datkim	Dat	Kim	john@example.co	1234567890	7 days	2023-01-08
CucBach	Cuc	Bach	jane@gmail.com	0987654321	12 Months	2026-8-01
ThanhToan	Thanh	Toan	alice@fpt.edu.vn	5551234567	Lifetime	2199-01-01
DinhChi	Dinh	Chi	aaadasd@fpt.edu	5551244467	Lifetime	2199-01-01
LongNhat	Long	Nhat	alas@fpt.edu.vn	551234567	Lifetime	2199-01-01
DatNguyen	Dat	Nguyen	emo@gmail.com	0123123123	12 Months	2026-07-21
TuDoi	Tu	Doi	d	dd	1 Months	2025-08-21
HungNguyen	Hung	Nguyen	aloalo@gmail.co	123123123123	3 Months	2025-10-20

Search and Edit Trainer

Search by Name:

Username	First Name	Last Name	Email	Phone	Trainer ID	Height	Weight
johndoe	John	Doe	john@example.	1234567890	1	180	80
janesmith	Jane	Smith	jane@gmail.co	0987654321	2	180	80
alicejohnson	Alice	Johnson	alice@fpt.edu.vi	5551234567	3	180	80
hoangthong	thong	hoang	john@example.	1234567890	1	180	80

Manage Subscription Plans

Subscription Plans, Durations (days) and Fees

1 Months:	30	days	1000000	VND
3 Months:	90	days	2800000	VND
6 Months:	180	days	5500000	VND
12 Months:	365	days	10000000	VND
36 Months:	1200	days	25000000	VND
Lifetime:	400000	days	100000000	VND
a:	123	days	123	VND
7 days:	7	days	200000	VND

Add New Subscription Plan

Plan Name: Duration (days): Fee (VND):

Some of the GUI in Admin

Revenue Calculation and Reports

Revenue Summary

Total Revenue: 344,800,000 VND

Total Members: 10

Revenue by Subscription Plan

Plan	Revenue (VND)	Members
1 Months	2,000,000	2
12 Months	40,000,000	4
Lifetime	300,000,000	3
3 Months	2,800,000	1

Export to CSV

Save Report (Default)

Back to Menu

Top Active Members

Ranked by attendance frequency

Rank	Name	Username	Attendance Coui	Last Attendance
#1	Cuc Bach	CucBach	6 days	07 Nov 2025
#2	Dat Kim	datkim	5 days	04 Nov 2025
#3	Thanh Toan	ThanhToan	4 days	07 Nov 2025
#4	Long Nhat	LongNhat	3 days	04 Nov 2025
#5	Tu Doi	TuDoi	3 days	04 Nov 2025
#6	Dinh Chi	DinhChi	2 days	04 Jul 2025
#7	Dat Nguyen	DatNguyen	2 days	04 Jul 2025
#8	Anh Anh	anhanh	2 days	05 Nov 2025

Add Attendance Day & Export Report

Select Member:

Dat Kim (datkim) ▼

Select Attendance Date:

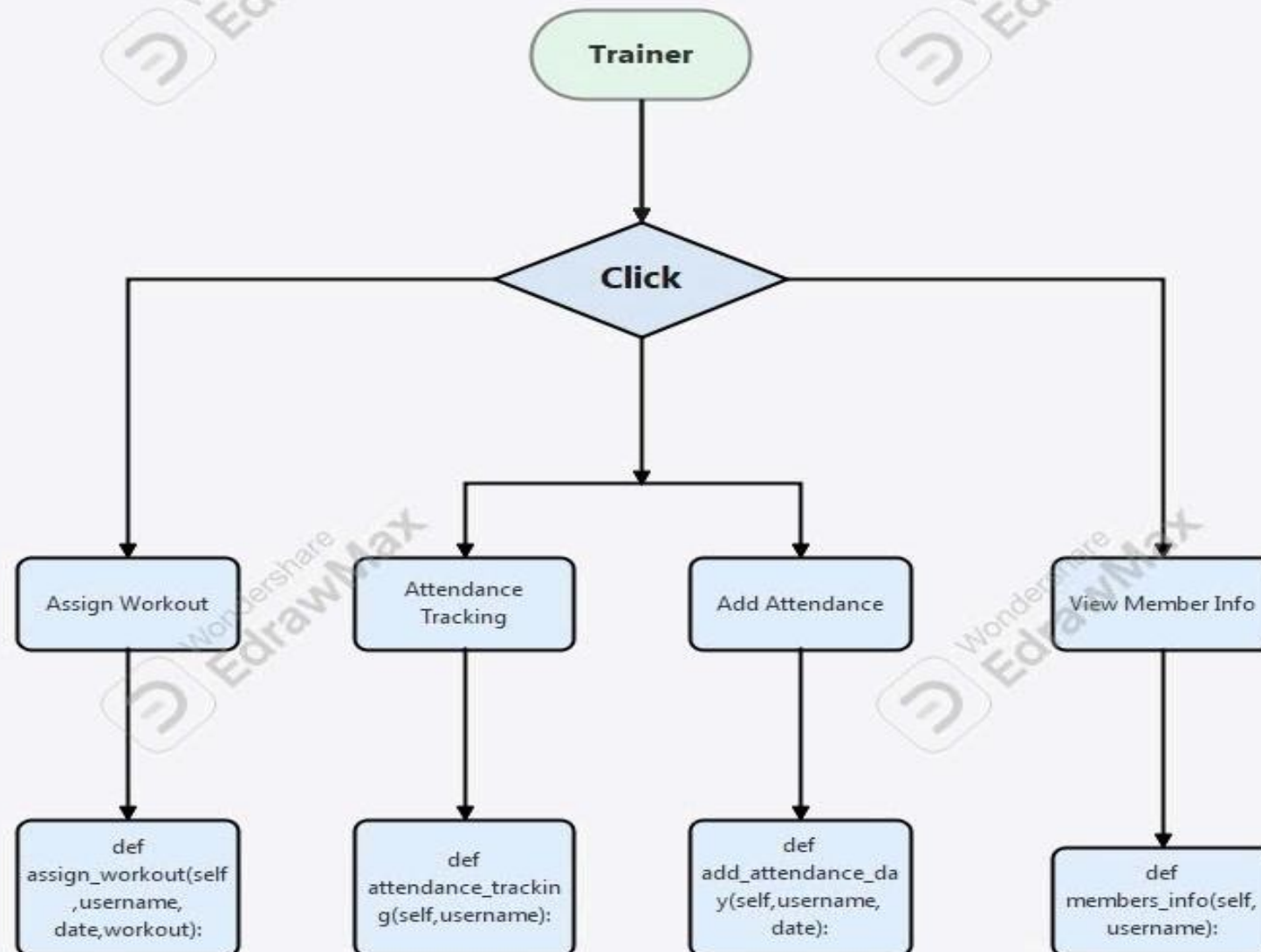
2025-11-12 ▼

Add Attendance

Export Attendance Report (CSV)

Back to Menu

Gym Management System Functions (Trainer)



TRAINER WINDOW

Gym Management System - Trainer



Assign Workout

Attendance Tracking

Add Attendance

View Member Info

Progress Tracking

Exit



Some of the GUI in Trainer

Attendance Tracking

Select Date:

Member:

Search:

No members attended on 2025-11-10

Assign Workout to Member

Select Member:

Select Workout Date:

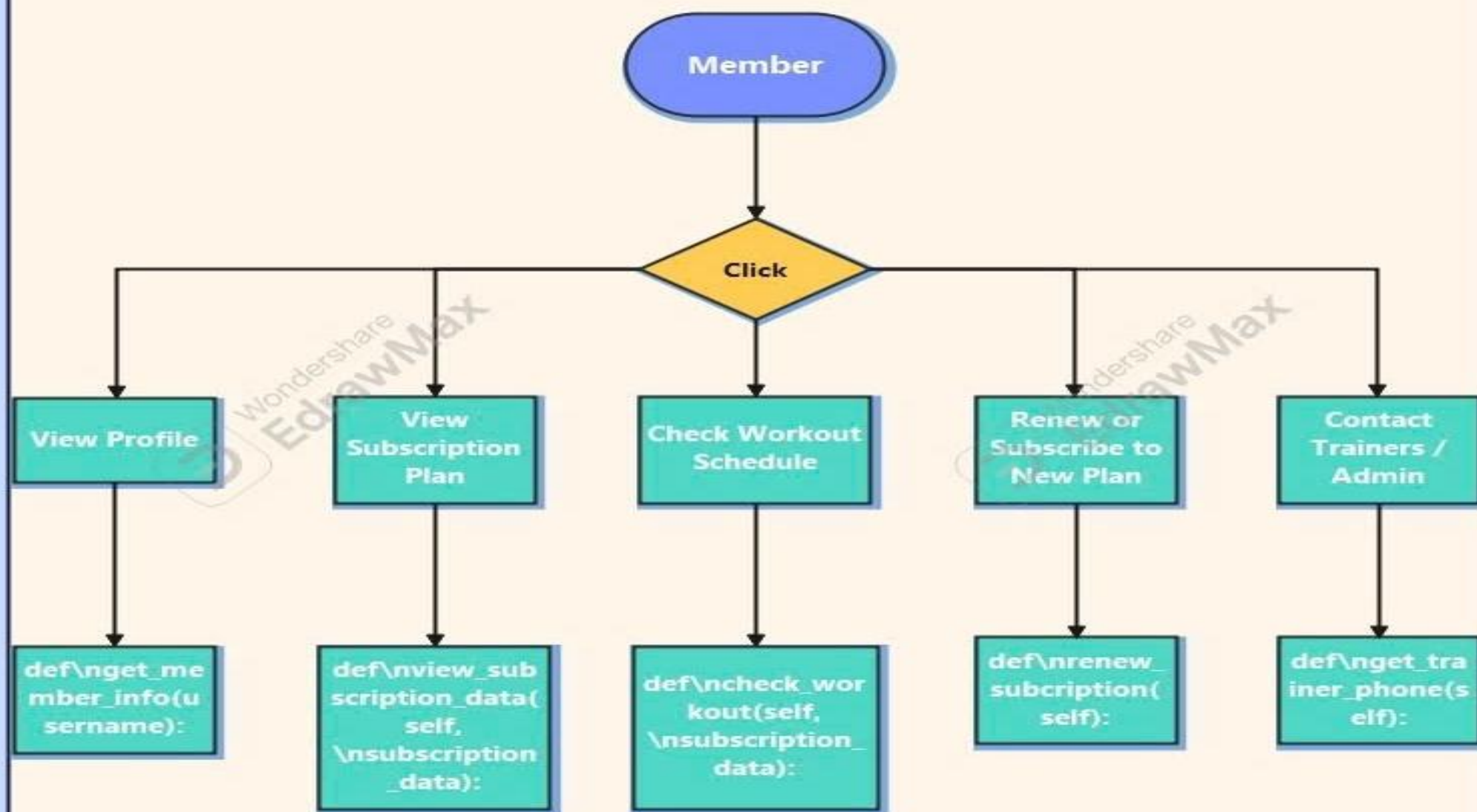
Workout Description:

Add Attendance Day & Export Report

Select Member:

Select Attendance Date:

Gym Management System Functions (Member)



MEMBER WINDOW

Gym Management System - Member

[View Profile](#)[Track Progress](#)[View Subscription Plan](#)[Check Workout Schedule](#)[Renew/Subscribe to New Plan](#)[Contact Trainers / Admin](#)[Exit](#)

Some of the GUI in Member

Member Profile: datkim

id: 1
f_name: Dat
l_name: Kim
gender: Male
phone: 1234567890
date_of_birth: 1990-01-01
joined_date: 2023-01-01
address: 123 Main St, City, Country
email: john@example.com
membership: 7 days
subscription_expiry: 2023-01-08
username: datkim
attendance_days: ['2025-07-03', '2025-07-04', '2025-07-01', '2025-10-22', '2025-11-04']
workout_days: [['2025-07-03', '100 Push-up, 10 jump rope'], ['2025-07-04', '50 Squat, 20 Burpees'], ['2025-07-24', 'Chay 100km'], ['2025-11-07', 'aaa']]
trainer_username: hoangthong
body_stats: [{ 'date': '2025-11-07', 'weight': 100.0, 'body_fat': 50.0 }, { 'date': '2025-11-07', 'weight': 122.0, 'body_fat': 69.0 }]

Close

Workout Days for datkim

Date: 2025-07-03
Workout: 100 Push-up, 10 jump rope

Date: 2025-07-04
Workout: 50 Squat, 20 Burpees

Date: 2025-07-24
Workout: Chay 100km

Date: 2025-11-07
Workout: aaa

Close

Your Subscription Plan

Plan: 1 Months
Duration: 30 days
Fee: 1,000,000 VND
Expiry Date: 2025-09-01
Trainer Username: johndoe

Close

Some of the GUI in Member

Renew or Subscribe to a New Plan

Your Current Plan

Plan: 1 Months

Expiry: 2025-09-01

Available Plans

1 Months
12 Months
3 Months
36 Months
6 Months
7 days
Lifetime
a
trial

Plan Details

Duration: 30 days

Price: 1,000,000 VND

Close

Confirm



Contact Trainer / Admin

Trainer's Phone:

1234567890

Admin's Phone:

0977889999



THANK YOU
FOR
LISTENING